

Unit 5: Portfolio Optimisation & Risk Management

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Mean–Variance Framework

Portfolio Optimisation in R

Portfolio Returns Analysis

GARCH on Portfolio Returns

Rolling Forecasts & VaR Backtesting

Exercises

By the end of this unit you will be able to:

- 1 Explain the mean–variance framework and diversification
- 2 Construct a minimum-variance portfolio using **PortfolioAnalytics**
- 3 Compute portfolio returns from individual asset returns
- 4 Fit a GARCH model to portfolio returns and forecast volatility
- 5 Backtest VaR using rolling forecasts and interpret Kupiec and Christoffersen coverage tests

Companion script

Open **Unit-5-Portfolio-Optimization.R** — it downloads multi-asset data, optimises a portfolio and performs VaR backtesting.

Mean–Variance Framework

Markowitz Portfolio Theory

The mean–variance framework (1952)

For N assets with returns $\mathbf{r} = (r_1, \dots, r_N)'$:

- Expected return: $\boldsymbol{\mu} = E[\mathbf{r}]$
- Covariance matrix: $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{r})$
- Weights: $\mathbf{w}$ with $\sum_i w_i = 1$

Portfolio return and risk:

$$\mu_p = \mathbf{w}'\boldsymbol{\mu}, \quad \sigma_p^2 = \mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}$$

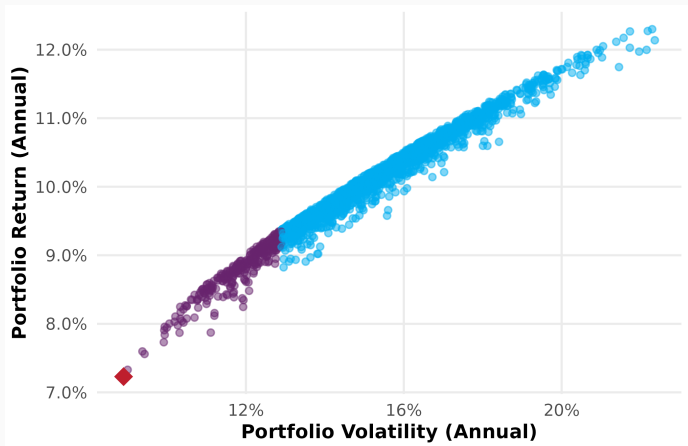
Diversification

Portfolio risk $\leq$ weighted sum of individual risks (unless perfectly correlated):

$$\sigma_p \leq \sum_i w_i \sigma_i$$

This is the **free lunch** of diversification.

The Efficient Frontier



Each point is a random portfolio. The upper-left boundary is the **efficient frontier** — portfolios with maximum return for a given risk level. The diamond marks the minimum-variance portfolio.

The **minimum variance portfolio** solves:

$$\min_w w' \Sigma w \quad \text{subject to} \quad w' \mathbf{1} = 1$$

Additional constraints:

- **Long-only:** $w_i \geq 0$
- **Box:** $l_i \leq w_i \leq u_i$
- **Group:** sector exposure limits

Optimisation

A **quadratic program** (QP) with linear constraints. Solved efficiently by dedicated QP solvers like OSQP.

Portfolio Optimisation in R

Five-step process:

- 1 Define: `portfolio.spec/assets()`
- 2 Constrain: `add.constraint()`
- 3 Objective: `add.objective()`
- 4 Optimise: `optimize.portfolio()`
- 5 Extract: read `$weights`

Constraint	Type
weight_sum	$\sum w_i = 1$
box	$l_i \leq w_i \leq u_i$
group	Sector limits

Objective	Name
Min risk	"StdDev"
Max return	"mean"
Max Sharpe	"mean"/"StdDev"

Setting Up and Solving

Requires: `library(PortfolioAnalytics)`; uses `%>%` from `dplyr/magrittr`.

```
init <- portfolio.spec(assets = symbols) %>%  
  add.constraint(type = "weight_sum", min_sum = 0.99, max_sum = 1.01) %>%  
  add.constraint(type = "box", min = 0, max = 1) %>%  
  add.objective(type = "risk", name = "StdDev")  
  
opt <- optimize.portfolio(R = returns, portfolio = init,  
  optimize_method = "osqp")
```

Extract results:

`opt$weights` — optimal allocation

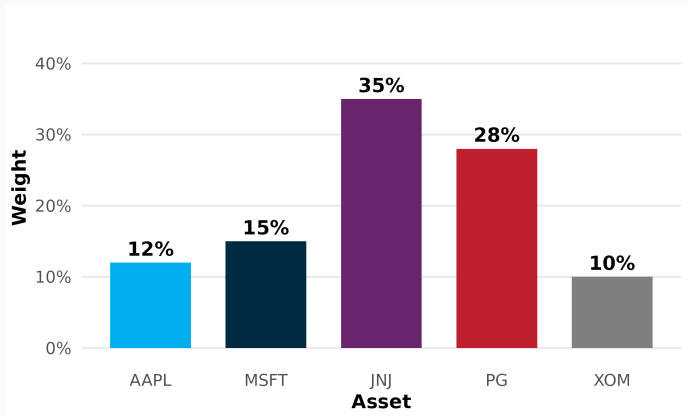
`opt$objective_measures` — achieved risk/return

Data format

Returns matrix: rows = dates, columns = assets.

Use `pivot_wider()` from long format.

Optimal Weights



A typical minimum-variance allocation concentrates in low-volatility, low-correlation assets (here JNJ). High-volatility tech stocks receive smaller weights.

Portfolio Returns Analysis

With weights $\mathbf{w}$ and individual returns $r_{i,t}$:

$$r_{p,t} = \sum_{i=1}^N w_i r_{i,t}$$

In R:

```
returns_long: cols date, symbol, return; weights_df: cols  
symbol, weight  
port_ret <- returns_long %>%  
  left_join(weights_df, by = "symbol") %>%  
  group_by(date) %>%  
  summarise(ret = sum(return * weight))
```

Next steps

Treat the portfolio return series as a univariate time series. Apply all tools from Units 1–4: visualisation, ARIMA, GARCH, VaR.

GARCH on Portfolio Returns

Portfolio returns inherit stylised facts from their constituents: volatility clustering, fat tails, leverage effects.

```
spec <- ugarchspec(  
  variance.model = list(model = "gjrgARCH",  
    garchOrder = c(1,1)),  
  mean.model = list(armaOrder = c(1,0),  
    include.mean = TRUE),  
  distribution.model = "std")  
  
fit <- ugarchfit(spec, data = port_ret)
```

Why GJR-GARCH?

Equity portfolios typically exhibit the leverage effect. Test whether **gamma1** is significant to confirm asymmetry.

Rolling Forecasts & VaR Backtesting

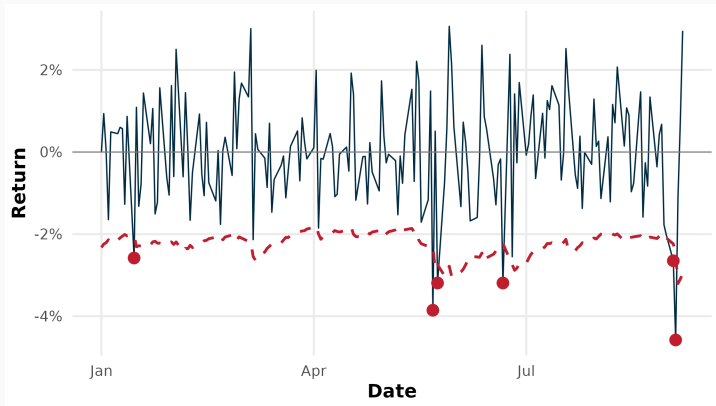
Rolling Forecast with `ugarchroll`

Convert tibble to xts: `port_xts <- xts(port_ret$ret, order.by = port_ret$date)`

```
roll <- ugarchroll(spec = spec, data = port_xts,  
  n.ahead = 1, forecast.length = 252,  
  refit.every = 50, refit.window = "moving",  
  calculate.VaR = TRUE, VaR.alpha = c(0.01, 0.05))
```

Parameter	Meaning
<code>forecast.length</code>	Out-of-sample window ($252 \approx 1$ year)
<code>refit.every</code>	Re-estimate every 50 observations
<code>refit.window</code>	"moving" or "recursive"
<code>VaR.alpha</code>	Tail probabilities (1%, 5%)

VaR Backtest



The black line shows actual portfolio returns. The dashed red line is the rolling 95% VaR threshold. Red dots mark **violations** — days when the actual loss exceeded VaR. For a well-calibrated model, violations should occur $\approx 5\%$ of the time.

Key question: Does the model's VaR accurately capture tail risk?

```
report(roll, type = "VaR", VaR.alpha = 0.05, conf.level = 0.95)
```

Kupiec's test:

- H_0 : violation rate = α
- Tests **unconditional coverage**
- Rejects if too many or too few violations

Christoffersen's test:

- H_0 : violations are independent
- Tests **conditional coverage**
- Rejects if violations cluster

A good model passes *both* tests at the chosen confidence level.

Exercises

Open `Unit-5-Portfolio-Optimization.R` – the script provides data download and optimisation examples before each exercise.

- ❶ **Data:** (a) Download daily prices for 5 stocks; (b) compute log returns; (c) reshape to wide format with `pivot_wider(names_from, values_from)`
- ❷ **Optimise:** (a) Define a min-variance portfolio with long-only constraints; (b) solve with `osqp`; (c) plot optimal weights
- ❸ **Portfolio returns:** (a) Join weights to returns with `left_join`; (b) compute weighted sum per date; (c) plot returns and squared returns
- ❹ **GARCH:** (a) Fit GJR-GARCH(1,1); (b) extract coefficients with `coef(fit)`; (c) check if `gamma1` is significant
- ❺ **Backtest:** (a) Run 252-day rolling VaR with `ugarchroll`; (b) use `report(roll, type="VaR")`; (c) compare violation rate to expected 5%

Hints: Convert to `xts` for `ugarchroll()`. Expected violations: $0.05 \times 252 \approx 13$.

You can now:

- Explain mean–variance optimisation and the benefit of diversification
- Construct and solve a minimum-variance portfolio with **PortfolioAnalytics**
- Compute portfolio returns from individual asset returns
- Fit GARCH models to portfolio returns and forecast volatility
- Perform rolling VaR backtests with **ugarchroll**
- Interpret Kupiec and Christoffersen coverage tests

Next: Unit 6 — Wrap-up & Resources